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ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2017-2018

DURATION: 3 Hours

FULL MARKS: 150

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.

There are 8 (eight) questions. Answer any 6 (six) of them.

Figures in the right margin indicate marks.

1. a) Explain the advantages and disadvantages of using a direct mapped cache instead of an 8-way set associative cache. 5
 b) A computer system uses 16-bit memory addresses. It has a 2K-byte cache organized in a direct-mapped manner with 64 bytes per cache block. Assume that the size of each memory word is 1 byte. 6+4
 - i. Calculate the number of bits in each of the tag, index, and offset fields of the memory address.
 - ii. Below is a sequence of two binary memory addresses in the order they are used to reference memory. Assume that the cache is initially empty. For each reference, write down the tag and index bits and indicate whether that reference is a hit or a miss

0000 0000 1000 0000, 0000 0000 1001 0000
- c) Repeat question 1(b), if the cache is organized as a 2-way set-associative cache that uses the LRU replacement algorithm. 6+4
2. a) Define Horizontal and vertical organization of micro instruction. Why is the WMFC step needed when reading from or writing to the main memory? 4+4
 b) Suppose a processor is using the single bus structure for internal datapath. Write the sequence of control steps required for each of the following instructions. 4+4
 - i. *Add (R3), R1*
 - ii. *Add R3, R1*
- c) Briefly describe hardwired control and microprogrammed control with proper diagram. 9
3. a) What do you mean by memory interleaving? How can the memory interleaving increase higher average utilization of memory? 7
 b) What is the advantage of using a write-back cache instead of a write-through cache? You find that it would be very inexpensive to implement small, direct mapped cache of 32K bytes with an access time of 30 ns. However, the hit rate would be only about 50%. If the main memory access time is 60 ns, does it make sense to implement the cache? 5+4
 c) The average memory access time for a microprocessor with 1 level of cache is 2.4 clock cycles 9
 If data is present and valid in the cache, it can be found in 1 clock cycle. If data is not found in the cache, 80 clock cycles are needed to get it from off-chip memory. Designers are trying to improve the average memory access time to obtain a 65% improvement in average memory access time, and are considering adding a 2nd level of cache on-chip. This second level of cache could be accessed in 6 clock cycles. The addition of this cache does not affect the first level cache's access patterns or hit times. Off-chip accesses would still require 80 additional CCs. To obtain the desired speedup, how often must data be found in the 2nd level cache?

4. a) What is virtual memory? Explain the concept of virtual memory address translation scheme with paging using necessary diagrams. 2+6
- b) What is TLB? Explain (in 4-5 sentences) why TLB is needed for virtual-to-physical address translation. 2+5
- c) Explain TLB hit and TLB miss with proper example and diagram. 10
5. a) Briefly explain the difference between page fault and TLB miss with proper example. 7
- b) What do you mean by RAID? List the characteristics of each RAID level with diagram. 8
- c) Write short note on read and write mechanism of magnetic disk. 7
- d) Consider a machine with 64-bit addresses and an 8KB page size. How many bits are required to represent a page number? 3
6. a) Consider the following sequence of instructions being processed on the pipelined 5-stage RISC processor 6x3
- Load R4, #100(R2)*
Add R5, R2, R3
Subtract R6, R4, R5
And R7, R2, R5
- i. Identify all the data dependencies in the above instruction sequence. For each dependency, indicate the two instructions and the register that causes the dependency.
- ii. Assume that the pipeline does not use operand forwarding. Also assume that the only sources of pipeline stalls are the data hazards. Draw a diagram that represents instruction flow through the pipeline during each clock cycle. How long does it take for the instruction sequence to complete?
- iii. Now, assume that the pipeline uses operand forwarding. There are separate forwarding paths from the outputs of stage-3 and stage-4 to the input of stage-3. Draw a diagram that represents the flow of instructions through the pipeline during each clock cycle. Indicate operand forwarding by arrows.
- b) What is dynamic branch prediction? Draw the finite state machine for a 2-bit prediction scheme. 2+5
7. a) "Pipelining does not result in individual instructions being executed faster; rather, it is the throughput that increases" Explain this with necessary example. 7
- b) Why shouldn't we use a million pipeline stages if an operation can be divided up into a million steps, even if we can keep the pipeline full? 5
- c) Briefly describe the different functional units of basic linear pipeline architecture. 6
- d) Consider the execution of a program of 15000 instructions by a linear pipeline processor with a clock rate of 25MHz. Assume that the instruction pipeline has 5 stages and that one instruction is issued per clock cycle. The penalties due to branch instructions and out-of-sequence executions are ignored 3+4
- i. Calculate the speedup factor as compared with non-pipelined processor
- ii. What are the efficiency and throughput of this pipelined processor
8. a) What do you mean by Direct Memory Access? Briefly explain how DMA controller can transfer a block of data from an external device to the processor? 2+6
- b) Write short notes on the following. 3x3
- i. Cycle Stealing
- ii. Memory-mapped I/O
- iii. Interrupt-service routine
- c) One way of receiving a response from a device is to poll it periodically about whether it has any additional information. What is wrong with this technique? Explain the alternative. 2+6

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WINTER SEMESTER, 2018-2019

DURATION: 3 Hours

FULL MARKS: 150

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.

There are **8 (eight)** questions. Answer any **6 (six)** of them.

Figures in the right margin indicate marks.

1. a) Write down the basic differences between the followings with suitable examples: 6
 - i. Computer organization and computer architecture
 - ii. Computer structure and computer function
- b) On the IAS, appositely describe the process that the CPU must undertake stepwise to read a value from memory and to write a value to memory in terms of what is put into the MAR, MBR, address bus, data bus, and control bus with necessary figures. 10
- c) A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. Ignore penalties due to branch instructions and out-of-sequence executions. 9
 - i. What is the speedup of this processor for this program compared to a non-pipelined processor?
 - ii. What is throughput (in MIPS) of the pipelined processor?
2. a) The instruction execution may involve several operations and depends on the nature of the instruction. Draw an instruction cycle state diagram with the interrupts which is required to comply in program execution cycle. Briefly describe all the states with a suitable example. 10
- b) What happens when a check bit rather than a data bit is in error? How many check bits are needed if the Hamming error correction code is used to detect single bit errors in a 1024-bit data word? Justify your answer. 7
- c) Consider a computer system that contains an I/O module controlling a simple keyboard/printer teletype (an electromechanical device that can be used to send and receive typed messages through various communications channels). The following registers are contained in the processor and connected directly to the system bus: 8

INPR: Input Register, 8 bits

OUTR: Output Register, 8 bits

FGI: Input Flag, 1 bit

FGO: Output Flag, 1 bit

IEN: Interrupt Enable, 1 bit

Keystroke input from the teletype and printer output to the teletype are controlled by the I/O module. The teletype is able to encode an alphanumeric symbol to an 8-bit word and decode an 8-bit word into an alphanumeric symbol. Now answer the following questions:

 - i. Describe how the processor, using the first four registers listed in this problem, can achieve I/O with the teletype.
 - ii. Describe how the function can be performed more efficiently by also employing IEN.
3. a) Define logical cache and physical cache. Comparatively distinguish them with necessary figures. 8

- b) Suppose that the processor has access to two levels of memory. Level 1 contains 1000 words and has an access time of $0.01 \mu\text{s}$; level 2 contains 100,000 words and has an access time of $0.1 \mu\text{s}$. Assume that if a word to be accessed is in level 1, then the processor accesses it directly. If it is in level 2, then the word is first transferred to level 1 and then accessed by the processor. For simplicity, we ignore the time required for the processor to determine whether the word is in level 1 or level 2. If 95% of the memory accesses are found in level 1, what is the average time to access a word? 5
- c) Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine. 12
- How is a 16-bit memory address divided into tag, line number, and byte number?
 - Into what line would bytes with each of the following addresses be stored?
 0001 0001 0001 1011
 1100 0011 0011 0100
 1101 0000 0001 1101
 1010 1010 1010 1010
 - Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?
 - How many total bytes of memory can be stored in the cache?
 - Why the tag is also stored in the cache?
4. a) Magnetic disk is the foundation of external memory on virtually all computer systems. Briefly describe how data are recorded on and later retrieved from the disk with necessary figures. 8
- b) Write short notes on following terms: 5
- Data striping
 - RAID 10
 - CD-R
 - MZR
 - Pit
- c) Consider a single-platter disk with the following parameters: rotation speed: 7200 rpm; number of tracks on one side of platter: 30,000; number of sectors per track: 600; seek time: one ms for every hundred tracks traversed. Let the disk receive a request to access a random sector on a random track and assume the disk head starts at track 0. 12
- What is the average seek time?
 - What is the average rotational latency?
 - What is the transfer time for a sector?
 - What is the total average time to satisfy a request?
5. a) The processor needs to save information stored in several registers in system stack to resume the current program at the point of interrupt. When interrupt processing is complete, the saved register values are retrieved from the stack and restored to the registers. Now draw several figures to explain the above mentioned scenario with a portion of main memory and different registers of processor. 7
- b) To implement interrupt driven I/O, respective design issues are raised. Mention those design issues and also briefly describe different techniques to solve them. 10
- c) What is I/O channel? How many different types of I/O channels are there? Describe them with respective figures. 8
6. a) Many instruction sets contain the instruction NOOP, meaning no operation, which has no effect on the processor state other than incrementing the program counter. Suggest some uses of this instruction. 5
- b) Compare one-, two-, and three- address machines by writing programs to compute the following expression for each of the three machines. 12

$$X = (A + B * C) / (D - E * F)$$

- c) For the following data structure (figure 1), draw the big-endian and little-endian layouts, using the values given as comment: 8

```

struct {
    short i;      //0x1112
    short j;      //0x1314
    short k;      //0x1516
    short l;      //0x1718
} s3;

```

Figure 1: A data structure

7. a) An instruction format defines the layout of the bits of an instruction. The design of an instruction format is a complex art, and an amazing variety of designs have been implemented. There are several design issues in this context. Briefly describe those issues mentioning different trade-offs. 8
- b) A sharp contrast to the instruction set of the PDP-8 is that of the PDP-10 which was designed to be a large-scale time-shared system, with an emphasis on making the system easy to program. Briefly describe different design principles employed in designing the instruction set of PDP-10. 5
- c) Given the following memory values and a one- address machine with an accumulator, what values do the following instructions load into the accumulator? 12
- Word 20 contains 40.
 - Word 30 contains 50.
 - Word 40 contains 60.
 - Word 50 contains 70.
- i. LOAD IMMEDIATE 20
 - ii. LOAD DIRECT 20
 - iii. LOAD INDIRECT 20
 - iv. LOAD IMMEDIATE 30
 - v. LOAD DIRECT 30
 - vi. LOAD INDIRECT 30
8. a) With the earliest computers, the programmer interacted directly with the computer hardware without OS. These early systems presented two main problems. Briefly describe those problems with necessary examples. 5
- b) Comparatively describe partitioning, paging and segmentation in memory management. Among them, segmentation has a number of advantages to the programmer over a non-segmented address space. Briefly remarks on those advantages. 10
- c) List and briefly explain various ways in which an instruction pipeline can deal with conditional branch instructions. 10

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SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2019-2020

DURATION: 1 Hour

FULL MARKS: 50

CSE 4305: Computer Organization and Architecture

General Instructions

- Write your Name, Student-ID and Course Code on the top of the first page. Maintain a serial number on the Top-right corner of each page.
- Answer all the questions. Figures in the right margin indicate marks.
- Sit in proper position and maintain the environment as per the Guidelines.
- No examinee is allowed to scan the file unless 1 hour is finished.
- For any circumstances, follow the instructions of the invigilator.

- 1 What is the difference between *micro-operation* and *micro-instruction*? Write the sequence of events (micro-operations) of "*interrupt cycle*" symbolically employing common processor registers with appropriate data flow diagram. 3+4
+3
- 2 Many processor designs include a register or set of registers, often known as the program status word (PSW) or flags, that contain status information. These flags typically contain condition codes plus other status information resulted from the immediately last operation. If the last operation was a subtraction operation between two operands, A and B containing 11110000 and 00010100 (*i.e.* A-B), what would be the value of any three of the following flags? 3

i. Carry	iv. Sign
ii. Zero	v. Even Parity
iii. Overflow	vi. Equal
- 3 Briefly describe different kinds of data hazards in any instruction pipeline system. (Note: Appropriate examples from each type are highly appreciable). 5
- 4 The Intel 8088 consists of a bus interface unit (BIU) and an execution unit (EU), which form a 2-stage pipeline. The BIU fetches instructions into a 4-byte instruction queue. The BIU also participates in address calculations, fetches operands, and writes results in memory as requested by the EU. If the bus is free and there is no outstanding requests and branch instructions, then answer the following questions: 2+3
+3
 - i. If the tasks done by the both units take equal time, by what factor does pipeline improve the performance of the 8088?
 - ii. Draw the timing diagram of instruction pipeline operation for 5 instructions.
 - iii. If there are 100 instructions in a program to be executed by the 8088, calculate the total time required to complete this program availing the instruction pipeline advancement.

In a 16-bit processor, at a certain time of the execution of instructions from main memory, following instructions are appeared (figure a):

Memory address	Content
200	XXMM
201	YYYY
202	WXYZ

Figure a: Contents of memory

Where XX denotes the opcode that instructs to load content to the register AC (accumulator), MM signifies the mode field, YYYY contains the decimal value 500 and WXYZ indicates the next instruction. For a specific mode, register R1 might be used containing the value 400. Assume that location 399 contains 999, location 400 contains 1000 and onwards accordingly. Determine the **effective address** and the **operand** to be loaded for any two of the following address modes represented by MM:

- Direct
- Indirect
- PC Relative
- Register
- Register Indirect

6 Convert the following high-level language statement to corresponding machine language program using one-address instruction format:

$$Y = (A \times B) + (C \times D) + E$$

7 Briefly describe the operation of the TLB and Cache to fetch any appropriate page in virtual memory system following the depiction given in figure b:

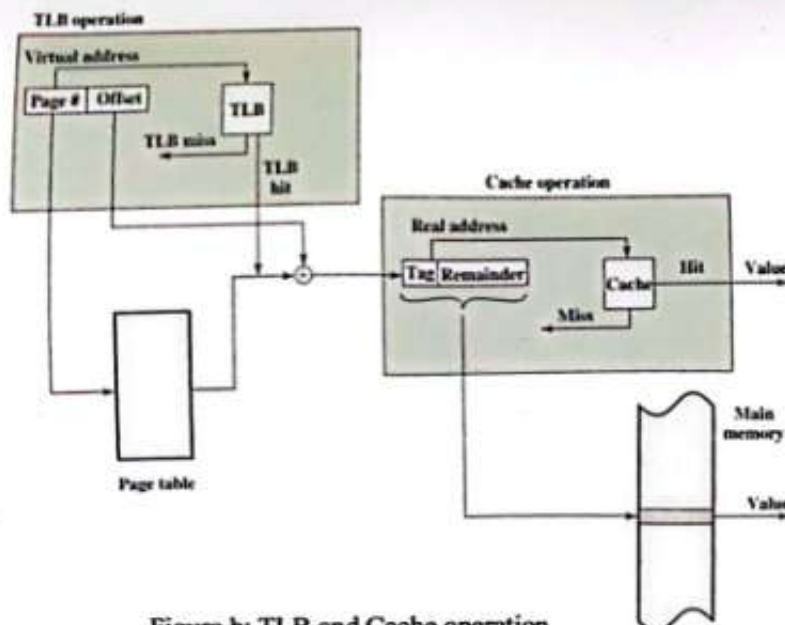


Figure b: TLB and Cache operation

8 What are the design issues to implement an interrupt driven I/O if there are multiple I/O devices? Mention the name of the appropriate approaches to overcome those issues.

- 9 Consider a magnetic disk drive with 10 surfaces, 600 tracks per surface, and 72 sectors per track. Sector size is 512 B. The average seek time is 10 ms, the track-to-track access time is 1.5 ms, and the drive rotates at 3600 rpm. Successive tracks in a cylinder can be read without head movement.
- What is the disk capacity?
 - What is the average access time? Assume this file is stored in successive sectors and tracks of successive cylinders start at sector 0, track 0 of cylinder i .
 - Estimate the time required to transfer a 5-MB file.

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SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2021-2022

DURATION: 3 HOURS

FULL MARKS: 150

CSE 4305: Computer Organization and Architecture**Programmable calculators are not allowed. Do not write anything on the question paper.**Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) It is a common practice to use a symbolic representation of machine instructions to write a machine-language program. Consider the following high-level language instruction: 2+2+3+8
(CO2)
(PO1)
- $$X = \frac{(A + B \times C)}{D - E \times F} \times (G + \frac{H}{I})$$
- What are the common sources of these operands?
 - What are the basic elements of a machine instruction?
 - Draw an instruction cycle state diagram showing all the steps involved in instruction execution
 - Write two separate programs using two- and one-address machine instruction formats respectively to compute the above expression. Values of given operands should not be changed during execution.
- b) A *procedure* is the most significant invention in the evolution of programming languages. It is an independent computer program that is embedded within a bigger program. The procedure may be invoked or called at any time within the program. An example program where procedures are called multiple times during its execution is illustrated in Figure 1. 5+5
(CO3)
(PO2)

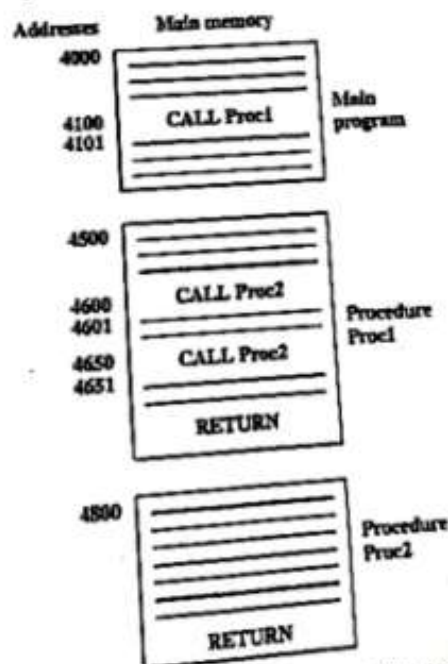


Figure 1: A program with procedure calls.

- Draw an execution sequence diagram illustrating how the procedures are called at different addresses during program execution.
- Based on the given scenario, demonstrate how a processor employs a stack data structure to store return addresses and implement nested procedure calls.

2. In a uniprogramming system, the main memory is divided into two sections: one for the operating system and one for the currently executing program. The latter section of this multiprogramming system's memory is partitioned to accommodate multiple processes. Suppose a process references five pages, A, B, C, D, and E, and the following sequence of virtual page numbers is observed on a computer with virtual memory:

A; B; C; D; A; B; E; A; B; C; D; E

- a) What is process? Why is its state considered important in multiprogramming? Draw a figure illustrating the life cycle of a process. 5
(CO3)
(PO2)
10
- b) Why would the main memory of a multiprogramming system such as the preceding example be expanded hypothetically? Discuss briefly the common methods utilized by operating systems to increase main memory with required figures. (CO3)
(PO2)
10
- c) Employ each replacement algorithm to compute the total number of page transfers during this sequence of references, beginning with an empty main memory that contains four-page frames. Demonstrate the intermediate phases of the calculation, taking into account any necessary assumptions. (CO3)
(PO2)
3. a) In a typical instruction style, the address field(s) are quite small. However, we prefer to access a large number of locations in main memory or virtual memory. Various addressing strategies have been utilized to achieve this objective. 2+5+5+8
(CO2)
(PO1)
- How does the processor determine the addressing mode of an instruction's operand(s)?
 - In displacement addressing mode, how does the CPU calculate the effective address (EA)? Which variations does this mode entail? Introduce them briefly.
 - Compare post- and pre-indexing addressing mode with the relevant figures.
 - If an instruction's address field contains the decimal value 14, determine the EA of the corresponding operand for all conceivable basic addressing modes. Assume a mandatory variable in any field that requires one and also discuss any anomalies, if any.
- b) Various techniques can be used to predict whether a branch will be taken or not. Among them dynamic branch prediction approaches depend on the execution history. Briefly explain those approaches with appropriate figures. 5
(CO5)
(PO1)
4. a) The ALU of a computer system can add two input registers and logically complement the bits of either register, but it doesn't have any built-in logic to perform subtraction. The integers must be stored in two's complement format in those registers. Assume R1 and R2 are the addresses of those registers. 4+5+5+6
(CO5)
(PO1)
- Write a program of machine instructions to perform subtraction operation between these two registers.
 - Draw the data flow diagram (i.e., a sequence of events) to perform the same subtraction operation.
 - Make a list the micro-operations that the associated control unit must do in order to reformulate the event sequence you have previously generated.
 - If the registers contain 11110000 and 0010100 respectively what would be the value of the following flags: Carry, Zero, Overflow, Sign, Even Parity, Half-Carry
- b) Explain "cycle stealing" with necessary figure. (CO
(PO

5. a) To the concept of instruction execution, we must recognize that, in fact, an instruction has a number of stages. Such a pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has six stages, and instructions are issued at a rate of one per clock cycle. 4+3+3+5
(CO5)
(PO1)
- i. Draw a flow chart of six stage "CPU instruction pipeline" depicting the logics necessary for branches and interrupts
 - ii. Ignoring penalties due to branch instructions and out-of-sequence executions what is the speedup of this processor for this program compared to a nonpipelined processor? Show the intermediate calculating steps.
 - iii. What is the pipelined processor's throughput (in MIPS) with the same assumptions?
 - iv. What is pipeline bubble? Provide some instances demonstrating how this event can occur in a variety of ways under the aforementioned circumstances, along with the required figures.
- b) Assume a disk drive comprises of four drives with a 200 GB capacity each. 4+6
(CO4)
(PO1)
- i. Draw a diagram illustrating the mapping between logical and physical disk space for such disk drive.
 - ii. What is the available data storage capacity for each of the RAID levels 0, 1, 3, 4, 5, and 6?
6. a) The issue with programmed I/O is that the CPU must wait a considerable amount of time for the I/O module in question to be prepared for data receiving or transmission. A different method for the processor to accept interrupts from I/O modules in order to complete such data communications. 2+6+6+6
(CO6)
(PO1)
- i. What role does the I/O module play in this strategy?
 - ii. The occurrence of an interrupt initiates a series of hardware and software processor activities. With an appropriate flow diagram explain how such interrupt is processed to complete an I/O operation in this way.
 - iii. Implementing Interrupt-driven I/O entailed a number of design issues. Discuss these concerns and their variants briefly.
 - iv. What is *interrupt controller*? Draw a diagram of a computer system illustrating how I/O devices can be connected to a processor using interrupt controller(s).
- b) The DMA mechanism can be configured in numerous ways to regulate bus access. Design some potential configurations for this aspect. 5
(CO6)
(PO1)

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WINTER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4305: Computer Organization and Architecture

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 Answer all 5 (five) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Categorize various cache types according to their location in relation to the processor, Memory Management Unit (MMU), and main memory. Compare their operational process with the required figures. 5
(CO1)
(PO1)
 - b) Consider a 16-bit data 0101000000111001 stored in the main memory. Generate a Hamming SEC-DED code for it that demonstrates each step in between. Additionally, demonstrate that the code accurately detects an error in data bit 5. 10
(CO1)
(PO1)
 - c) In pipeline planning, a special technique known as branch prediction can be utilized for predicting whether a branch will be taken to avoid branch hazard. Provide a brief explanation of the various branch predictions techniques along with relevant figures and examples. 10
(CO1)
(PO1)
2. a) Magnetic read-write operations refer to the process by which data is written to and subsequently read from a magnetic disk. Illustrate these mechanisms with the use of suitable figures. Additionally, briefly describe the formatting and organization of data on the platter of a magnetic disk that illustrates the disk's data structure. 8 + 7
(CO1)
(PO1)
 - b) Differentiate between the following pairs: 5 × 2
(CO1)
(PO1)
 - i. CLV and CAV
 - ii. Strip and stripe
 - iii. Hamming code and Hamming distance
 - iv. Page and frame
 - v. PCI and PCIe
3. a) There has been an upward trend of increasing component sophistication and complexity as computer systems have developed. The I/O function is the one place where this is most obvious. Provide an appropriate example to summarize the I/O Function's evolution process. 5
(CO1)
(PO1)
 - b) A computer and an external device are connected via an I/O module. Give a brief explanation of an I/O module's main features along with the required figures. 5
(CO1)
(PO1)
 - c) Two addressing modes—memory mapped and isolated—are available when the CPU, main memory, and I/O share a common bus. Explain the differences between isolated and memory-mapped I/O using relevant examples. 5
(CO1)
(PO1)
 - d) The majority of virtual memory schemes use a unique cache for page table entries, sometimes referred to as a Translation Lookaside Buffer (TLB), to lower the memory access time. Nevertheless, interaction between the virtual memory mechanism and the conventional cache system is required. 5 × 2
(CO1)
(PO1)
 - i. Illustrate the operational stages of a TLB using a flowchart.
 - ii. Show the interaction between a TLB and a cache using a figure.

4. A part of main memory content of a 16-bit architecture-based computer system is presented in Figure 1 showing different instructions and data in different locations. It has an instruction set having the following instructions with their corresponding opcodes (4-bit):

- Load AC from memory (0001)
- Store AC to memory (0010)
- ADD to AC from memory (0101)

300	1 9 4 0
301	5 9 4 1
302	2 9 4 1
...	
...	
940	0 0 0 3
941	0 0 0 2

Figure 1: A part of main memory for Question 4.

- a) Convert the instructions stored in memory locations 300-302 into instructions that have 1, 2, and 3 operands. With appropriate assumption(s), variables could be used in place of physical addresses. (CO1) (PO1) 10
- b) Considering the diagram depicted in Figure 2, generate the time sequence of micro-operations and the corresponding control signals to demonstrate the operation of the control unit. The control signal terminations are denoted as C_i and represented by a circular symbols in the figure. (CO3) (PO2) 15

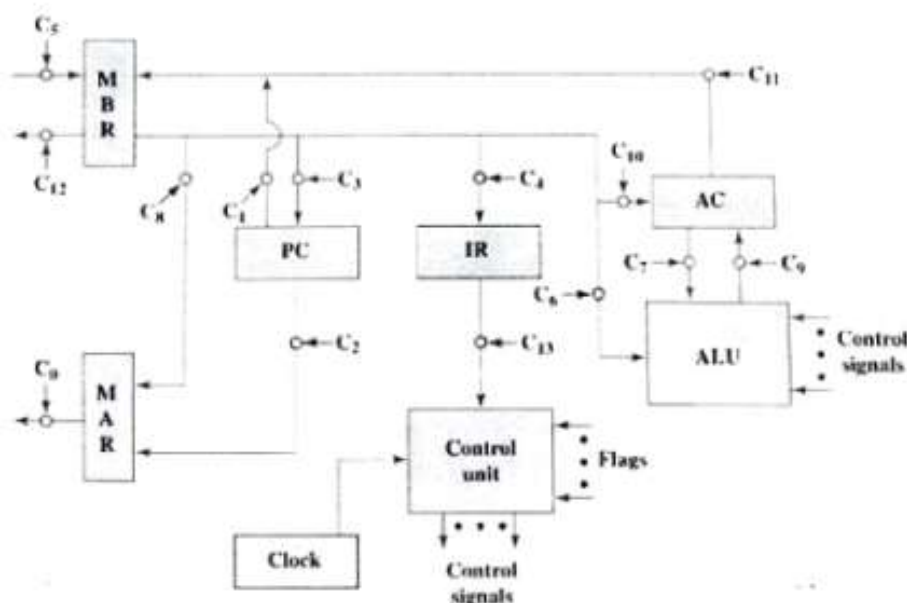


Figure 2: Data paths and control signals for Question 4.b)

- c) Demonstrate the events caused in Question 4.b) by utilizing CPU Registers - IR, PC, AC, MAR, and MBR. Illustrate how their contents are modified over time sequence for all three instructions. (CO3) (PO2) 15
- d) Verify the possibility of different types of hazards in the given scenario when implementing a pipeline strategy. Provide specific examples of these circumstances and propose effective solutions to address these problems. (CO3) (PO2) 10

5. a) The C structure provided in Code Snippet 1 includes a number of data types. The comments show the content of the corresponding variables. Illustrate the addressing and byte order of big-endian and little-endian layouts on a 64-bit architecture.
- 10
(CO1)
(PO1)

```

1 struct{
2     short i;      //0x1112
3     int j;        //0x13141516
4     double k;     //0x1718192021222324
5     int l;        //
6     char m[7];    //'A','B','C','D','E','F','G'
7 } s;

```

Code Snippet 1: A C structure for Question 5.

- b) Consider a byte addressable 16-bit microprocessor where a certain portion of main memory contains data and instruction as depicted in Figure 3. The memory contents from location 401 increase sequentially following the locations 399 – 400.
- 15
(CO1)
(PO1)

200	Load AC	Mode
201	500	
202	Next Instruction	
...		
...		
399	999	
400	1000	

Figure 3: A part of main memory for Question 5.b)

If source register, $R1 = 400$, base register = 100, and index register = 100, determine the Effective Address (EA) and the content of AC after executing instruction at address 200 for the following addressing modes:

- Direct
- Indirect
- Register
- Register indirect
- Immediate
- PC relative
- Displacement
- Indexing